

SUNAO SUGIYAMA

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CONTACT INFORMATION

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RESEARCH INTERESTS

Theoretical and Observational cosmology: *large-scale structure of the Universe, gravitational weak/micro lensing, primordial black hole*

COLLABORATIONS

Subaru HSC weak lensing working group, member (2021-present, **co-chair** since 2024Nov)
Dark Enrgy Survey (DES), member (2024-present), LSST member(2024-present), Roman HLIS (2024-present)

POSITIONS

Current	Project researcher , Kavli IPMU, Chiba, Japan	Feb. 2026 – present
Past	Postdoctoral researcher , University of Pennsylvania, Philadelphia, the United States Mentor: Prof. Bhuvnesh Jain	Sep. 2023 – Jan. 2026
	JSPS Overseas Research Fellow , University of Pennsylvania, Philadelphia, the United States	Sep. 2023 – Aug. 2025
	Project researcher , Kavli IPMU, Chiba, Japan Mentor: Prof. Masahiro Takada, funded partially by beyond AI	Apr. 2023 – Aug. 2023
	JSPS Research Fellowships for Young Scientists , Kavli IPMU, Chiba, Japan	Apr. 2021 – Mar. 2023
Educations	University of Tokyo, Tokyo, Japan , Ph.D. course in Physics Dissertation: “ <i>Joint cosmology analyses using gravitational weak lensing data from Subaru Hyper Suprime-Cam</i> ” Supervisor: Prof. Masahiro Takada	Apr. 2020 – Mar. 2023
	University of Tokyo, Tokyo, Japan , M.S. in Physics Dissertation: “ <i>Validation of cosmological analysis based on perturbation theory for wide-field galaxy survey</i> ” Supervisor: Prof. Masahiro Takada	Apr. 2018 – Mar. 2020
	University of Tokyo, Tokyo, Japan , B.A. in Physics	Apr. 2014 – Mar. 2018

GRANT & AWARDS

Grant-in-Aid for JSPS Research Fellows (DC2), Japan Society for the Promotion of Science, Apr. 2021 – Mar. 2023

The School of Science Encouragement Award (Doctoral program), University of Tokyo, the School of Science, Mar. 2023

WINGS IGPEES, course completion, Sep. 2018 – Mar. 2023

TEACHING

Collaborative coding: git and github, **CD3 symposium** 2023, Kavli IPMU

Coadvised students: Rafael C. H. Gomes (a PhD student at UPenn since 2023), Kyle Miller (a visiting student at UPenn, since 2024), Noriaki Nakasawa (a master student at the University of Nagoya, 2022)

PROFESSIONAL ACTIVITIES

Society	The Astronomical Society of Japan (ASJ), 2018 – present The Physical Society of Japan (JPS), 2022 – present
Service	IPMU weekly lunch seminar (co-organizer), 2019 – 2021 HSC weaklensing mini workshop (organizer), Aug. 2022 Sesto 2025 - Tracing Cosmic Evolution with Galaxy Clusters V (SOC), 2025
Referee	International Journal of Modern Physics D The Astrophysical Journal Astronomy & Astrophysics Journal of Cosmology and Astroparticle Physics
Computing	Developer of fft-extended-source , fastnc , dark emulator as a part of Dark Quest Project I can work with C, C++, Python, HSC pipeline (for image analysis)
Accepted Observation	PI of Definitive search for PBH dark matter in the multiverse cosmology with HSC co-I of Survey of M31 eclipsing binaries: Toward a 1% distance measurement

PUBLIC ENGAGEMENT AND OUTREACH

Apperence on *Cosmic Front, Primordial Black Hole: the master key of the Universe (title in Japanese)* produced by NHK, 2021

Interviewed by Quanta Magazine on *Clashing Cosmic Numbers Challenge Our Best Theory of the Universe*, 2024

Interviewed by Asahi Shimbun on *Breakdown in the standard model of the Universe? An analysis using precise dark matter “map” (title in Japanese)*, 2024

Interviewed by Jonathan O’Callaghan for *“Is This a Moon-Sized Primordial Black Hole Adrift in the Milky Way?” Scientific American*, 2026.

The up-to-date list of publication available at [ADS](#).

* = Author list alphabetized

First-author papers or co-authored papers with significant contributions

1. **Sugiyama, Sunao**, M. Takada, N. Yasuda, and N. Tominaga. Microlensing constraints on Primordial Black Hole abundance with Subaru Hyper Suprime-Cam observations of Andromeda. *arXiv e-prints*, arXiv:2602.05840, [February 2026:arXiv:2602.05840](#)
2. R. C. H. Gomes, K. Miller, **Sugiyama, Sunao**, et al. Tidal alignment and tidal torquing modeling for the cosmic shear three-point correlation function and mass aperture skewness. *arXiv e-prints*, arXiv:2601.09133, [January 2026:arXiv:2601.09133](#)
3. **Sugiyama, Sunao**, R. C. H. Gomes, and B. Jain. Cosmology from a joint analysis of second- and third-order shear statistics with Subaru Hyper Suprime-Cam Year 3 data. *Phys. Rev. D*, 112(12):123545, [December 2025:123545](#)
4. R. C. H. Gomes, **Sugiyama, S.**, B. Jain, et al. Dark Energy Survey Year 3 Results: Cosmological constraints from second- and third-order shear statistics. *Phys. Rev. D*, 112(12):123515, [December 2025:123515](#)
5. R. C. H. Gomes, **Sugiyama, S.**, B. Jain, et al. Cosmology with second- and third-order shear statistics for the Dark Energy Survey: Methods and simulated analysis. *Phys. Rev. D*, 112(12):123514, [December 2025:123514](#)
6. **Sugiyama, Sunao** and M. Park. Data compression with noise suppression for inference under noisy covariance. *Phys. Rev. D*, 112(12):123505, [December 2025:123505](#)
7. **Sugiyama, Sunao**, R. C. H. Gomes, and M. Jarvis. Fast modeling of the shear three-point correlation function. *arXiv e-prints*, arXiv:2407.01798, [July 2024:arXiv:2407.01798](#)
8. X. Li, T. Zhang, **Sugiyama, Sunao**, et al. Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear two-point correlation functions. *Phys. Rev. D*, 108(12):123518, [December 2023:123518](#)
9. H. Miyatake, **Sugiyama, Sunao**, M. Takada, et al. Hyper Suprime-Cam Year 3 results: Cosmology from galaxy clustering and weak lensing with HSC and SDSS using the emulator based halo model. *Phys. Rev. D*, 108(12):123517, [December 2023:123517](#)
10. **Sugiyama, Sunao**, H. Miyatake, S. More, et al. Hyper Suprime-Cam Year 3 results: Cosmology from galaxy clustering and weak lensing with HSC and SDSS using the minimal bias model. *Phys. Rev. D*, 108(12):123521, [December 2023:123521](#)
11. R. Dalal, X. Li, A. Nicola, et al. Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear power spectra. *Phys. Rev. D*, 108(12):123519, [December 2023:123519](#)
12. S. More, **Sugiyama, Sunao**, H. Miyatake, et al. Hyper Suprime-Cam Year 3 results: Measurements of clustering of SDSS-BOSS galaxies, galaxy-galaxy lensing, and cosmic shear. *Phys. Rev. D*, 108(12):123520, [December 2023:123520](#)
13. **Sugiyama, Sunao**, M. Takada, and A. Kusenko. Possible evidence of axion stars in HSC and OGLE microlensing events. *Physics Letters B*, 840:137891, [May 2023:137891](#)
14. H. Miyatake, **Sugiyama, Sunao**, M. Takada, et al. Cosmological inference from an emulator based halo model. II. Joint analysis of galaxy-galaxy weak lensing and galaxy clustering from HSC-Y1 and SDSS. *Phys. Rev. D*, 106(8):083520, [October 2022:083520](#)
15. H. Miyatake, Y. Kobayashi, M. Takada, et al. Cosmological inference from an emulator based halo model. I. Validation tests with HSC and SDSS mock catalogs. *Phys. Rev. D*, 106(8):083519, [October 2022:083519](#)
16. **Sugiyama, Sunao**. Fast Fourier Transformation Based Evaluation of Microlensing Magnification with Extended Source. *ApJ*, 937(2):63, [October 2022:63](#)
17. **Sugiyama, Sunao**, M. Takada, H. Miyatake, et al. HSC Year 1 cosmology results with the minimal bias method: HSC $\times$ BOSS galaxy-galaxy weak lensing and BOSS galaxy clustering. *Phys. Rev. D*, 105(12):123537, [June 2022:123537](#)

18. **Sugiyama, Sunao**, V. Takhistov, E. Vitagliano, et al. Testing stochastic gravitational wave signals from primordial black holes with optical telescopes. *Physics Letters B*, 814:136097, [March 2021:136097](#)
19. *A. Kusenkov, M. Sasaki, **Sugiyama, Sunao**, et al. Exploring Primordial Black Holes from the Multiverse with Optical Telescopes. *Phys. Rev. Lett.*, 125(18):181304, [October 2020:181304](#)
20. **Sugiyama, Sunao**, M. Takada, Y. Kobayashi, et al. Validating a minimal galaxy bias method for cosmological parameter inference using HSC-SDSS mock catalogs. *Phys. Rev. D*, 102(8):083520, [October 2020:083520](#)
21. **Sugiyama, Sunao**, T. Kurita, and M. Takada. On the wave optics effect on primordial black hole constraints from optical microlensing search. *MNRAS*, 493(3):3632–3641, [April 2020:3632–3641](#)
22. H. Niikura, M. Takada, N. Yasuda, et al. Microlensing constraints on primordial black holes with Subaru/HSC Andromeda observations. *Nature Astronomy*, 3:524–534, [April 2019:524–534](#)

co-authored papers

23. D. Rana, S. More, H. Miyatake, et al. Hyper Suprime-Cam Y3 results: Photo-z bias calibration with lensing shear ratios and cosmological constraints from cosmic shear. *Phys. Rev. D*, 113(6):063534, [March 2026:063534](#)
24. R. Terasawa, M. Takada, T. Kurita, and **Sugiyama, Sunao**. Late-time suppression of structure growth as a solution for the S8 tension. *Phys. Rev. D*, 112(8):083556, [October 2025:083556](#)
25. R. Terasawa, M. Takada, **Sugiyama, Sunao**, and T. Kurita. Testing small-scale modifications in the primordial power spectrum with Subaru HSC cosmic shear, primary CMB, and CMB lensing data. *Phys. Rev. D*, 112(4):043514, [August 2025:043514](#)
26. T. Zhang, X. Li, **Sugiyama, Sunao**, et al. Cosmology and Source Redshift Constraints from Galaxy Clustering and Tomographic Weak Lensing with HSC Y3 and SDSS using the Point-Mass Correction Model. *arXiv e-prints*, arXiv:2507.01386, [July 2025:arXiv:2507.01386](#)
27. T. Zhang, **Sugiyama, Sunao**, S. More, et al. Modelling Galaxy Clustering and Tomographic Galaxy-Galaxy Lensing with HSC Y3 and SDSS using the Point-Mass Correction Model and Redshift Self-Calibration. *arXiv e-prints*, arXiv:2507.01377, [July 2025:arXiv:2507.01377](#)
28. R. Terasawa, X. Li, M. Takada, et al. Exploring the baryonic effect signature in the Hyper Suprime-Cam Year 3 cosmic shear two-point correlations on small scales: The S8 tension remains present. *Phys. Rev. D*, 111(6):063509, [March 2025:063509](#)
29. K.-F. Chen, I.-N. Chiu, M. Oguri, et al. Weak-Lensing Shear-Selected Galaxy Clusters from the Hyper Suprime-Cam Subaru Strategic Program: I. Cluster Catalog, Selection Function and Mass - Observable Relation. *The Open Journal of Astrophysics*, 8:2, [January 2025:2](#)
30. T. Sunayama, H. Miyatake, **Sugiyama, Sunao**, et al. Optical cluster cosmology with SDSS redMaPPer clusters and HSC-Y3 lensing measurements. *Phys. Rev. D*, 110(8):083511, [October 2024:083511](#)
31. I.-N. Chiu, K.-F. Chen, M. Oguri, et al. Weak-Lensing Shear-Selected Galaxy Clusters from the Hyper Suprime-Cam Subaru Strategic Program: II. Cosmological Constraints from the Cluster Abundance. *The Open Journal of Astrophysics*, 7:90, [October 2024:90](#)
32. J. Shi, T. Sunayama, T. Kurita, et al. The intrinsic alignment of galaxy clusters and impact of projection effects. *MNRAS*, 528(2):1487–1499, [February 2024:1487–1499](#)
33. T. Zhang, X. Li, R. Dalal, et al. A general framework for removing point-spread function additive systematics in cosmological weak lensing analysis. *MNRAS*, 525(2):2441–2471, [October 2023:2441–2471](#)
34. Y. Park, T. Sunayama, M. Takada, et al. Cluster cosmology with anisotropic boosts: validation of a novel forward modelling analysis and application on SDSS redMaPPer clusters. *MNRAS*, 518(4):5171–5189, [February 2023:5171–5189](#)

Other Articles

1. S. Sugiyama, M. Takada, and H. Miyatake. Weak lensing cosmology with subaru hsc data. *ASJ EUREKA*, 117(1):304–314, [May 2024:304–314](#)

Listing 20 selected talks out of 50 talks. See [here](#) for the full list of talks.

1. **Probing Primordial Black Hole Dark Matter with Subaru HSC Microlensing Observations of M31**, [IPNS Joint Experimental-Theoretical Cosmology Seminar](#), 2026, Jun., *Oral* (Invited Talk)
2. **Searching for Primordial Black Holes with Microlensing Data from Subaru HSC**, [ASIAA colloquium](#), 2026, Apr., *Oral* (Invited Talk)
3. **Observational Search for Primordial Black Holes Using Subaru/HSC Microlensing**, [Elucidating the Universe's Creation with Neutrinos based on International Science Collaborations](#), 2026, Mar., *Oral* (Invited Talk)
4. **Observational search of PBH**, [Dark matter and black holes](#), 2025, Dec., *Oral* (Invited Talk)
5. **Data Compression with Noise Suppression for Inference under Noisy Covariance**, Roman F2F meeting, 2025, Nov., *Oral*
6. **Cosmology from a joint analysis of second and third order shear statistics**, KICP seminar, 2025, Oct., *Oral*
7. **Cosmology with third-order shear statistics Applications to HSC and DES**, [Beyond-two-point Statistics Meet Survey Systematics](#), 2025, Sep., *Oral* (Invited Talk)
8. **Exploring Primordial Black Hole with Microlensing Data: Updates on Analysis Pipeline**, [Focus week on primordial black holes 2024](#), 2024, Nov., *Oral* (Invited Talk)
9. **Exploring Primordial Black Hole with Microlensing Data**, [Pacific conference](#), 2024, Aug., *Oral* (Invited Talk)
10. **Cosmology from Subaru HSC weak lensing Year 3 data**, [MIFA colloquium](#), 2024, May., *Oral* (Invited Talk)
11. **Cosmology from weak lensing three-point correlation function**, astro/cosmo seminar at CMU, 2024, Feb., *Oral*
12. **Cosmology from Subaru HSC weak lensing Year 3 data**, [Subaru Users Meeting FY2023](#), 2024, Jan., *Oral*
13. **HSC Y3 weak lensing cosmology results**, [CosmoPalooza](#), 2023, Oct., *Oral*
14. **HSC Year 3 Weak Lensing Cosmology Results**, [HSC webinar](#), 2023, Apr., *Oral*
15. **HSC Y3 cosmology results**, [CMB x LSS](#), 2023, Apr., *Oral* (Invited Talk)
16. **Collaborative coding: git and github**, [CD3 Opening Symposium](#), 2023, Apr., *Oral*
17. **Revealing the nature of dark matter with gravitational lensing: weak and microlensing**, [Colloquium at Osaka theoretical astrophysics group](#), 2022, Jul., *Oral* (Invited Talk)
18. **Exploring Primordial black hole with microlensing observation of Andromeda galaxy**, [Subaru Users Meeting 2021](#), 2022, Jan., *Oral*
19. **Joint analysis of galaxy-galaxy lensing and clustering at large scales from Subaru HSC and SDSS data**, [2021 Autumn Annual Meeting of ASJ](#), 2021, Sep., *Oral*
20. **Testing stochastic gravitational wave signals by PBH microlensing**, [4th KEK-PH + KEK-Cosmo Joint Lectures and Workshop on "Gravitational Wave"](#), 2020, Nov., *Oral* (Invited Talk)

PRESS RELEASES

[Primordial black holes and the search for dark matter from the multiverse](#), IPMU, 2020 Dec

[How to see the invisible: Using dark matter distribution to test our cosmological model](#), IPMU, 2024 Apr